

James Nickoli

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Education

McMaster University

B.A.Sc. in Honours Computer Science

Hamilton, Ontario, Canada

Sep 2021 – Apr 2026

Information and Communications Technology Council

Service Design Course

May 2022 – Jul 2022

- Used the Service Design process while working with UI/UX designers to create a healthcare app prototype

Work Experience

Big Viking Games

Co-op Software Developer

London, Ontario, Canada

May 2022 – Aug 2022

- Increased game management efficiency by translating internal tools from **PHP** to **React**.
- Upgraded in-game social systems built using **Java**, **MySQL**, and **Typescript**, enhancing play experience for thousands of live users.
- Strengthened player connections by discovering sources of long standing bugs via the debugging features of **Docker** containers.
- Collaborated in an **Agile** team using **Jira**, **Confluence**, and **Slack** to maintain live service games running on **Firebase** and **Redis**.

Programming Experience

McMaster Competitive Programming Team

Team Member

Oct 2021 – Feb 2023

- Represented McMaster in the International Collegiate Programming Contest using **C++** and **Python** as a member of the junior team in 2021-2022, and placed 3rd out of 26 contestants in the 2022-2023 tryouts, qualifying as a member of the leading team.

First Robotics Competition

Team 5024 programming team member

Sep 2016 – Jun 2021

- Wrote readable and reliable **Java** and **C++** code for a 120-pound robot while debugging software, firmware, and hardware within a 6 week development time frame.
- Used **GitHub**, and **Trello** to collaborate with the programming sub-team, and **Slack** to work with the broader team.

Projects

Collectibles Management App

github.com/collectiblescapstone/Kollec-App

- Helped to create a cross-platform collectibles tracking and trading app using **Next.js**, **Capacitor**, **Vercel**, and **Supabase**.
- Created a system to identify trading cards from a user's camera using **YOLO**, **OpenCV**, and **ONNX**, achieving average processing speeds of 0.16 seconds on desktop and 0.5 seconds on mobile.
- Used unit testing in a CI/CD pipeline and performed debugging on native Android, native iOS, and web platforms.

Multiplayer Mastermind

github.com/rsninja722/MasterMind

- Built a custom **Java** package for making GUI programs (github.com/rsninja722/gamej) using **AWT** and **Swing**, handling multi-threading, rendering that leverages the GPU, asset loading, camera systems, and mouse and keyboard input, also providing documentation and project templates.
- Defined a communication protocol and wrote a server using **Netty** that can connect users with different clients

Missile Command in Custom Game Engine

github.com/rsninja722/UbisoftNext2024

- Participated in Ubisoft's NEXT competition, using the provided line drawing API to create a 3D game engine in **C++**, implementing memory pools and an Entity Component System
- Benchmarked different custom **C++** implementations of vector/matrix math, using **Python** to visualize performance testing results
- Recreated the arcade game Missile Command with 3D graphics, and physics-based missiles

Personal Site

<https://rsninja.dev>

- Maintaining a personal site, using **Google Analytics** to monitor traffic, currently over 900 users have visited without advertising